Model Integration Methodology: Smart Room Energy Management System

This document outlines the detailed system architecture, data flow, feature engineering pipeline, and integration mechanisms implemented to deploy the hybrid machine learning model within the Smart Room Energy Management System. This section is structured for direct inclusion or adaptation in the project methodology report.

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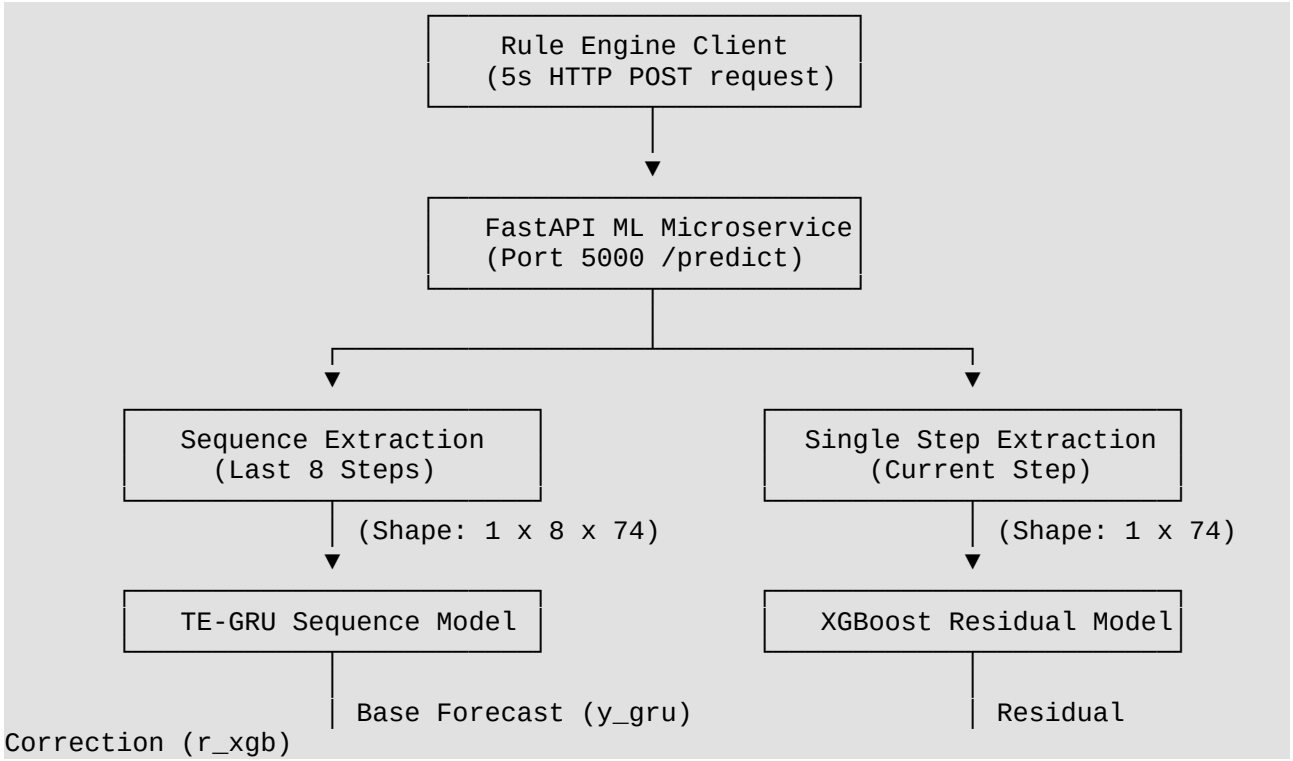
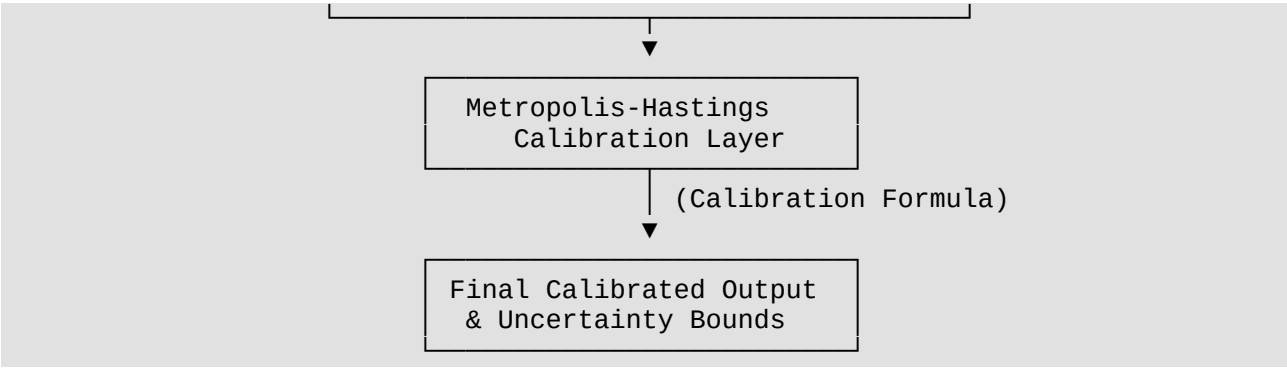
1. System Architecture & Model Pipeline

The prediction subsystem employs a hybrid, multi-stage machine learning architecture designed to forecast short-term energy demand (measured in Watt-hours, Wh) over a 5-minute Energy Demand Forecast Interval (EDFI). The system integrates three distinct analytical components:

1. **Temporal Encoding Gated Recurrent Unit (TE-GRU):** A recurrent neural network (RNN) designed to process sequential data. It evaluates a temporal window of the last **8 consecutive steps** (representing 40 minutes of continuous history) to capture sequential dependency, baseline trends, and thermal/occupancy inertia.
2. **Extreme Gradient Boosting (XGBoost) Residual Regressor:** A tree-based ensemble model that operates on the current single timestep. Its primary function is to predict and correct the residual error of the TE-GRU's temporal forecast using the immediate room state.
3. **Metropolis-Hastings (MH) Calibration & Uncertainty Estimation:** A Bayesian blending layer that weights the predictions of the GRU and XGBoost models, applying a calibrated adjustment formula. It also calculates safety margins (80% and 95% prediction intervals) representing the statistical uncertainty of the forecast.





The final calibrated energy prediction ($\hat{y}$) is calculated using the outputs of the TE-GRU ($\hat{y}*{\text{GRU}}$) and XGBoost ($\hat{r}*{\text{XGB}}$) models according to the following equation:

$$\hat{y} = \hat{y}*{\text{GRU}} + \alpha \cdot \hat{r}*{\text{XGB}} + \beta$$

Where:

- $\alpha = 1.091811$ (Scaling parameter for the residual correction)
- $\beta = -0.085535$ (Additive intercept adjustment)
- $Q_{95} = 2.0577$ (95% confidence interval bound)
- $Q_{80} = 0.8464$ (80% confidence interval bound)

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2. Feature Engineering Pipeline

The model does not operate on raw, isolated sensor readings. Instead, it utilizes an extensive feature engineering pipeline that transforms a raw 5-element telemetry vector into a **74-column feature matrix** representing the environmental, temporal, and historical state of the room.

For each of the 8 timesteps in the sequence window, the pipeline computes the following features:

2.1 Environmental Telemetry (4 Columns)

Direct measurements acquired from physical sensors:

- `temperature`: Room ambient temperature (DHT22 sensor, °C).
- `humidity`: Room relative humidity (DHT22 sensor, %).
- `lux`: Light intensity (BH1750 light sensor, lx).
- `occupancy`: Room occupancy count (estimated using ultrasonic and radar motion sensors).

2.2 Temporal & Calendar Indicators (6 Columns)

Time characteristics derived from the server-synchronized system clock:

- `hour`: Hour of the day (0 to 23).

- `minute`: Minute of the hour (0 to 59).
- `dayofweek`: Day of the week (0 to 6, where 0 is Monday).
- `month`: Month of the year (1 to 12).
- `day`: Day of the month (1 to 31).
- `is_weekend`: Binary flag (1 if the day is Saturday or Sunday; 0 otherwise).

2.3 Trigonometric Time Encodings (4 Columns)

Trigonometric transformations applied to periodic temporal features to preserve cyclical continuity (e.g., ensuring 23:59 is mathematically adjacent to 00:01):

- `hour_sin` / `hour_cos`: Sine and Cosine representation of the hour: $$\text{hour_sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad \text{hour_cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$
- `month_sin` / `month_cos`: Sine and Cosine representation of the month: $$\text{month_sin} = \sin\left(\frac{2\pi \cdot \text{month}}{12}\right), \quad \text{month_cos} = \cos\left(\frac{2\pi \cdot \text{month}}{12}\right)$$

2.4 Interaction Cross-Features (5 Columns)

Multiplicative combinations of live variables to capture non-linear relationships:

- `temp_x_humidity`: Temperature $\times$ Humidity
- `temp_x_occupancy`: Temperature $\times$ Occupancy
- `humidity_x_occupancy`: Humidity $\times$ Occupancy
- `lux_x_occupancy`: Light Level $\times$ Occupancy
- `hour_x_occupancy`: Hour $\times$ Occupancy

2.5 Historical Energy Lags & Statistics (22 Columns)

Temporal characteristics capturing past energy states. These are computed and duplicated under two separate namespace prefixes (`energy_` and `real time energy_`) to maintain compatibility with different model versions:

- **Lags:** `lag1`, `lag2`, `lag3` (representing actual energy consumed 5, 10, and 15 minutes ago, respectively).
- **Rolling Averages:** `roll3`, `roll6`, `roll12` (moving averages of energy consumption over the last 15, 30, and 60 minutes).
- **Rolling Volatility:** `std6`, `std12` (moving standard deviation of energy over the last 30 and 60 minutes).
- **Trends:** `trend3`, `trend6`, `trend12` (difference between the previous timestep and the energy level N steps prior).

2.6 Historical Sensor Lags & Statistics (33 Columns)

Lags, rolling averages, rolling standard deviations, and trends (matching the formulas above) are calculated for the environmental sensors:

- **Temperature history:** 11 columns (lags 1–3, rolling averages 3/6/12, rolling standard deviations 6/12, trends 3/6/12).
- **Humidity history:** 11 columns.
- **Occupancy history:** 11 columns.

Total Feature Count: $4 \text{ (raw)} + 6 \text{ (calendar)} + 4 \text{ (cyclical)} + 5 \text{ (interactions)} + 22 \text{ (energy lags)} + 33 \text{ (sensor lags)} = 74 \text{ columns}$.

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3. Data Hydration & Buffer Management

The ML service is designed as an isolated microservice that does not persist state across reboots. Thus, specialized hydration strategies are employed.

3.1 Cold Start Bootstrapping

When the FastAPI microservice initializes, the rolling memory buffer is hydrated with historical data to prevent Keras shape mismatches and mathematical calculation errors on initial prediction requests.

- **Mechanism:** The service attempts to read the last 60 records from a localized file (`mydatanew.csv`) containing training data.
- **In-Memory Buffer:** These records are loaded into a thread-safe `history_buffer` deque with a maximum capacity of 60.

3.2 Live Ingestion & MQTT Subscriber

As the system operates, the database and the in-memory buffer are updated with real-world sensor values:

- **Subscriber:** An independent background thread in the ML microservice runs an MQTT client subscribed to the `room/sensors` topic.
- **Ingestion:** Telemetry payloads containing cumulative energy readings and sensor data are normalized and appended to the in-memory buffer.
- **Eviction:** When a new reading is pushed, the oldest reading in the deque is evicted. Within 60 minutes of live operation, all initial bootstrap rows are replaced by actual real-time room data.

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4. API & Integration Protocol

The ML service runs as a headless REST API utilizing **Uvicorn** and **FastAPI** on port `5000`.

4.1 Prediction Request Schema

The rule engine issues HTTP `POST` requests to `/predict`. The body contains the immediate real-time environmental snapshot:

```
{
  "temperature_c": 30.5,
  "humidity": 61.2,
  "lux": 150.0,
  "occupancy": 3,
  "datetime_str": "2026-06-24T22:30:00"
}
```

Note: The current energy is omitted in the body, as that is the target variable. The historical energy lags for the prediction row are calculated using the previous step values stored in the in-memory buffer.

4.2 Prediction Response Schema

The service returns a flat JSON payload containing the predictions, bounds, and contributing model values:

```
{
  "predicted_energy_wh": 13.4,
  "upper_bound_energy_wh": 15.4577,
  "lower_bound_energy_wh": 11.3423,
  "predicted_energy_range_wh": [11.3423, 15.4577],
  "energy_unit": "Wh",
  "tegru_raw_wh": 12.6476,
  "xgb_residual_wh": 0.7675,
  "peak_demand": 5.0,
  "buffer_size": 60,
  "timestamp": "2026-06-24T22:30:00.123456Z",
  "source": "fastapi-tegru-xgb-mh-db"
}
```

4.3 Control Loop vs. Dashboard Polling

A design challenge was coordinating the polling frequency:

- **Dashboard Polling (Real-Time):** The rule engine calls the `/predict` endpoint every **5 seconds** and publishes the result to `room/ml/predictions` on the MQTT broker. This ensures that the user interface has a high-fidelity, real-time prediction graph.
- **Control Decisions (5-Minute Boundary):** Physical relay actuation decisions are restricted to 5-minute intervals. The rule engine uses the latest prediction cached in memory when the 5-minute decision boundary is reached.

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5. Decision Logic & Actuation

The predicted energy (referred to as the Energy Demand Forecast Interval, or EDFI) is fed directly into the system's control logic to actuate the room's appliance relays.

5.1 The EDFI Threshold Rules

The system evaluates the room's energy mode based on the predicted energy load and the stability of the battery storage system:

Load Class	**Condition**	**Battery $\ge 24.5\text{V}$**	**Battery $\ge 24.0\text{V}$**	**Battery $\ge 23.5\text{V}$**	**Battery $< 23.5\text{V}$**
Peak	EDFI $\ge 25\text{ Wh}$	Mode A (All ON)	Mode B (HVAC+Frz)	Mode C (Frz only)	Mode C (Frz only)
Moderate	EDFI $\ge 10\text{ Wh}$	Mode B (HVAC+Frz)	Mode B (HVAC+Frz)	Mode C (Frz only)	Mode C (Frz only)
Baseline	EDFI $\ge 1\text{ Wh}$	Mode C (Frz only)	Mode C (Frz only)	Mode C (Frz only)	Mode C (Frz only)
Very Low	EDFI $< 1\text{ Wh}$	Mode C (Frz only)	Mode C (Frz only)	Mode C (Frz only)	Mode C (Frz only)

5.2 Battery Stability Filter

To prevent rapid relay switching (chattering) caused by transient voltage spikes or sags under load, the rule engine implements a temporal lag filter on the battery voltage:

- **Tracking:** The system tracks the last 3 battery voltage readings sampled at 30-second intervals (T_{now}, T_{-1}, T_{-2}).
- **Enforcement:** A mode can only be activated if all 3 historical readings in the stability queue are strictly above the threshold voltage.

5.3 Actuation Pathway

The rule engine does not actuate the physical relays directly. Instead, it publishes the target states (`relay_1`, `relay_2`, `relay_3`) and the selected `mode` to the `room/relays/state` topic on the MQTT broker. An ESP32 microcontroller connected to the relay board subscribes to this topic and performs the physical actuation.

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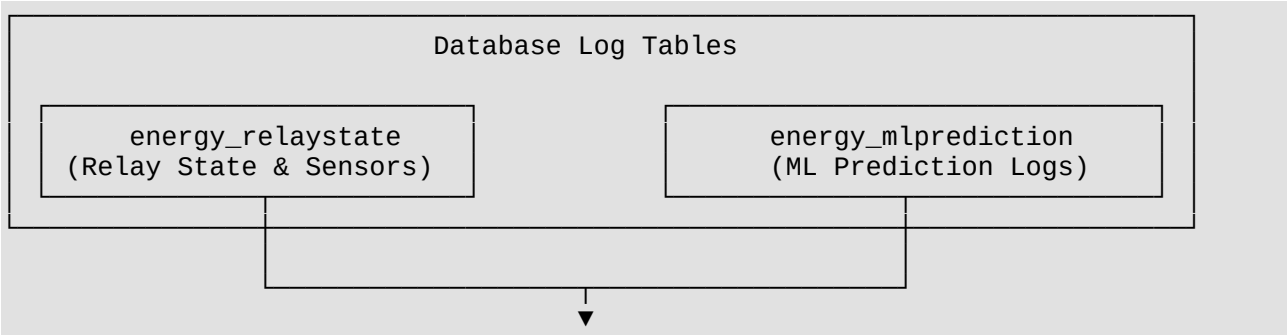
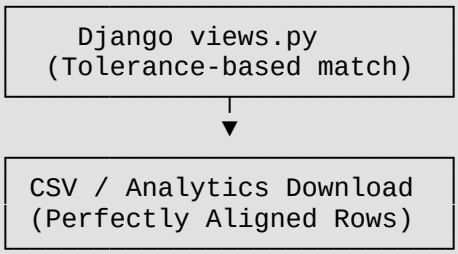
6. Database Logging & Alignment

All sensor readings, prediction outputs, and mode decisions are persisted to a SQLite database (`db.sqlite3`) managed by a Django backend.





6.1 Logging Desynchronization

- **Relay State Logging:** The Rule Engine evaluates decisions every 5 minutes and writes the room state immediately to the `energy_relaystate` database table.
- **Prediction Logging:** The MQTT Logger worker subscribes to `room/ml/predictions`, buffers the results, and flushes them to the `energy_mlprediction` table.
- **The Offset Challenge:** Because the prediction log is written asynchronously, the database timestamp for the prediction is offset by several seconds relative to the decision timestamp.

6.2 The Alignment Matching Algorithm

To align these records for post-hoc analysis and CSV downloads, the Django backend uses a tolerance-based future lookup window:

1. For each `RelayState` record at timestamp T_{decision}, the view queries the `MLPrediction` table.
2. It looks for the first prediction record with a timestamp $T_{\text{prediction}}$ that falls within a 6-minute window starting at the decision time: $$T_{\text{decision}} - 30\text{s} \le T_{\text{prediction}} \le T_{\text{decision}} + 5\text{m }30\text{s}$$
3. This 30-second skew tolerance handles minor system clock offsets, and the 6-minute upper bound ensures that decisions are not matched with far-future predictions in the event of logging gaps.
4. This ensures that every row in the exported analytics output matches the exact environment state, the forecast generated for that state, and the resulting relay mode decision.